

Microwave Imaging Framework

Detailed Project Report

End-to-end microwave breast imaging: acquisition → preprocessing → reconstruction → ROI → detection → characterization → confidence → reporting → auto validation

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Repository: github.com/Alwin1122/microwave | Branch: breawave-dev

Modules: 12/12 complete (core) | Unit tests: 101 passing

Important: Model / reconstruction output is **not a clinical diagnosis**. Results are research and educational estimates under stated assumptions and must not be used for medical decision-making.

1. Project overview

This desktop PySide6 application implements a full microwave imaging pipeline for breast-phantom / research datasets. It supports two primary input paths: University of Manitoba BMID frequency-domain cubes (already adipose-reference cleaned) and UG Touchstone .s2p measurements that require the full Module 2 Week 1–3 preprocessing brief. Downstream modules reconstruct images with DAS, DMAS, and DMAS-D4, localize a suspicious ROI, decide tumor candidacy, characterize the region in centimeters, score reconstruction confidence, visualize results, and export session reports. A production automation layer discovers datasets, classifies quality (pass/warning/fail/blocked), and writes dual guide/beginner reports.

1.1 Goals

- Load and validate measurement files (.mat, Touchstone .sNp, UM-BMID multi-scan cubes).
- Meet UG Module 2 preprocessing requirements for .s2p while keeping BMID _adi pass-through.
- Reconstruct and compare DAS / DMAS / DMAS-D4 with quality-based selection.
- Localize, refine, detect, characterize, and confidence-score suspicious regions.
- Eliminate manual batch checking via one-click / CLI / MCP auto validation.
- Produce reviewable Markdown/JSON/PNG/PDF documentation for guides and beginners.

1.2 End-to-end flow

Step	Modules	User action / output
Load	1	Choose .mat / BMID scan / .s2p
Preprocess	2 (+3 path)	BMID pass-through or .s2p Week 1–3
Reconstruct	3–5	DAS/DMAS/DMAS-D4 + beamformer pick
Localize & detect	6–8	ROI, refine, tumor candidate Yes/No
Characterize & score	9–10	Size/shape/intensity + overall confidence
Review & export	11–12	GUI plots + MD/JSON/PNG reports
Auto validate (optional)	Automation	Batch quality gates + dual reports

2. Two data paths (critical)

Path	Input	Module 2	Why
UM-BMID	fd_data_s21_adi.mat + md_list_*.mat	Pass-through (all none)	_adi already reference-cleaned; tumor/healthy from metadata
UG .s2p brief	Touchstone .s2p	Full Week 1–3 S21 pipeline	Official Module 2 PDFs describe this track

Large fd_data_*.mat cubes are gitignored; place them locally under project/datasets/. Companion metadata lists (md_list_*) may be present for scan labels and tumor ground truth.

3. Module catalog (detailed)

Module 1 — Data Acquisition and Validation [Completed]

Purpose: Load measurement data into MicrowaveDataset and validate integrity.

Input: File path / BMID cube

Output: Validated dataset + metadata

Handling logic: Extension checks; MATLAB/Touchstone parsers; BMID scan picker; frequency/S-parameter validation.

Module 2 — Signal Preprocessing [Completed]

Purpose: Clean S21 for .s2p; pass-through for BMID _adi.

Input: Dataset + config

Output: Filtered / windowed / time-domain variants + NRMSE report

Handling logic: Spike fix, filters, Hamming, IFFT, reference/clutter/normalize; conditional steps reported.

Module 3 — Frequency-to-Time Conversion [Completed]

Purpose: Provide time-domain signals for beamforming.

Input: Frequency-domain S21

Output: Time traces + time axis

Handling logic: IFFT with $dt = 1/(N \Delta f)$; used by reconstruction and Week-3 path.

Module 4 — Initial Breast Image Reconstruction [Completed]

Purpose: Form 2D microwave images of the FOV.

Input: Signals + antenna geometry

Output: DAS, DMAS, DMAS-D4 images

Handling logic: Shared grid/FOV/wave-speed; BMID-friendly geometry defaults.

Module 5 — Quality Assessment & Beamformer Selection [Completed]

Purpose: Score images and select the best reconstruction.

Input: Candidate images

Output: Selected image + metrics

Handling logic: SNR/SCR/CCR/contrast/FWHM; modes: quality, prefer/force DMAS-D4, tumor GT.

Module 6 — Suspicious Region Localization (ROI) [Completed]

Purpose: Find the most suspicious hotspot.

Input: Selected image

Output: Mask, bbox, centroid, score

Handling logic: Peak / off-center / tight modes; edge and size penalties.

Module 7 — ROI High-Resolution Reconstruction [Completed (core)]

Purpose: Re-image the ROI at higher resolution.

Input: ROI bbox + signals

Output: Refined local image

Handling logic: Finer local grid; compact-focus preference for stable refinement.

Module 8 — Tumor Detection (Candidate Decision) [Completed (core)]

Purpose: Decide if ROI is tumor-like or clutter-like.

Input: ROI features

Output: Yes/No + confidence + reason

Handling logic: Suspicion features with penalties for large edge-touching clutter.

Module 9 — Tumor Characterization [Completed (core)]

Purpose: Quantitative physical descriptors of the candidate.

Input: ROI + FOV spans

Output: cm size/shape/intensity metrics

Handling logic: Equivalent diameter, bbox, FWHM, compactness, local SCR, position.

Module 10 — Reconstruction Confidence Assessment [Completed (core)]

Purpose: One interpretable 0–1 reliability score.

Input: Metrics + candidate + localization

Output: Overall score + label + reasons

Handling logic: Weighted blend: image quality, detection, beamformer margin, localization, focus.

Module 11 — Visualization Dashboard [Completed (core)]

Purpose: Interactive review of the full workflow.

Input: Pipeline state

Output: Plots and tables

Handling logic: Acquisition / Preprocess / Reconstruct tabs; light Fusion theme.

Module 12 — Automated Report Generation [Completed]

Purpose: Export reviewable session + automation documentation.

Input: Session / batch context

Output: MD, JSON, PNG, PDF

Handling logic: Autosave under results/; Report button; dual guide/beginner automation reports.

4. Auto validation automation layer

A reusable service eliminates manual batch checking: discover files under configurable roots, classify metadata vs measurement, resolve `md_list_*` → `fd_data_*` pairs, auto-select BMID scans (default first tumor else first), run load→preprocess→reconstruction→ROI checks, classify with threshold profiles (strict / balanced / lenient), cache duplicate verifications, and write Report A + Report B.

4.1 Interfaces

- GUI: Acquisition tab → **Run Auto Validation** (background thread, log streaming, opens guide report).
- CLI: `python tools/run_auto_validation.py --roots datasets --profile balanced --max-files 5`
- MCP stdio: `automation/mcp_server.py` tools — `run_auto_validation`, `get_latest_auto_report`, `list_recommended_uploads`, `explain_dataset_status`

4.2 Quality gates (examples)

Gate	Balanced default
Finite output	Required
Min energy	1e-12
Min ROI area	4 px
Max localization error (if GT)	3.0 cm
Min SCR / SNR / confidence	2.0 / 0.5 / 0.35

5. Software architecture

Package	Role
<code>data_loader/</code>	Load/validate MATLAB, Touchstone, BMID
<code>preprocessing/</code>	Filters, calibration, Week 1–3 S21, handoff export
<code>reconstruction/</code>	IFFT, DAS/DMAS/DMAS-D4, ROI refine, auto-calibrate
<code>roi/</code>	ROI detection + tumor candidate
<code>quality/</code>	Metrics, beamformer select, characterization, confidence
<code>automation/</code>	Discover/resolve/policy/cache/classify/reports/MCP
<code>gui/</code>	PySide6 Acquisition / Preprocess / Reconstruct
<code>utils/</code>	Session reports, logging, exceptions
<code>tools/</code>	Headless CLI scripts
<code>tests/</code>	unittest suite (101 tests)

6. How to run

6.1 GUI

- `cd project && python main.py`
- Load dataset → Run preprocessing → Run reconstruction → inspect Details / ROI / confidence
- Optional: Acquisition → Run Auto Validation for batch screening
- Export session report via Report / Download Final Report (writes under results/)

6.2 Tests

```
python -m unittest discover -s tests -v
```

6.3 Auto validation CLI

```
python tools/run_auto_validation.py --roots datasets --profile balanced --max-files 5
```

7. Evidence from current workspace

The following evidence is taken from generated artifacts in `project/results/` at report build time. Counts and statuses reflect the latest auto-validation run available locally.

7.1 Latest guide report excerpt

```
- Generated: 2026-09-15 22:51
- Threshold profile: **lenient**
- Summary: Validated 1 target(s): 1 pass, 0 warning, 0 fail, 0 blocked.
## 1. Module-wise progress
| Module | Status | Purpose | Input | Output | Handling logic |
|---|---|---|---|---|---|
| 1 Acquisition | Completed | Load & validate measurements | `.mat` / `.snp` / BMID cube | `MicrowaveDataset` | Extension checks, BMID scan pick, metadata extract |
| 2 Preprocessing | Completed | Clean S21 (`.s2p`) or pass-through BMID | Dataset + config | Filtered/time-domain variants | Conditional spike/filter/Hamming/IFFT/ref/clutter |
| 3 Freq→Time | Completed | IFFT path for beamforming | Frequency S21 | Time signals | `dt=1/(NAf)` axis |
| 4 Reconstruction | Completed | Form image | Time/freq S + geometry | DAS/DMA5/DMA5-D4 images | Shared FOV/grid |
| 5 Quality select | Completed | Pick best beamformer | Candidate images | Selected image + scores | Quality / prefer / force / GT modes |
| 6 ROI localize | Completed | Find suspicious region | Selected image | Mask/bbox/centroid | Compactness + edge penalties |
| 7 ROI refine | Completed (core) | High-res ROI re-image | ROI bbox | Refined image | Local finer grid |
| 8 Detection | Completed (core) | Tumor-candidate decision | ROI features | Yes/No + confidence | Suspicion + clutter penalties |
| 9 Characterization | Completed (core) | Quantitative descriptors | ROI + FOV | cm size/shape/intensity | Physical pixel scales |
| 10 Confidence | Completed (core) | Unified reliability score | Metrics + candidate | 0-1 + label | Weighted components |
| 11 Visualization | Completed (core) | Interactive review | Pipeline state | Plots/tables | PySide6 tabs |
| 12 Reporting | Completed | Export session + automation docs | Context/batch | MD/JSON/PNG/PDF | Autosave + one-click automation |
#### Innovations beyond baseline
## 2. Current validation evidence (this run)
#### Pass / warn / fail counts
| Status | Count |
|---|---|
| pass | 1 |
| warning | 0 |
| fail | 0 |
| blocked | 0 |
#### Sample dataset statuses
| File | Scan | Status | Recommended | Key metric / reason |
|---|---|---|---|---|
```

7.2 GUI evidence map

- One-click starts at Acquisition → Run Auto Validation (beside Load dataset).
- Progress streams into the Acquisition Log panel as [Auto N%] messages.
- On completion: summary popup + automatic open of `latest_guide_progress_report.md`.
- Files saved under `project/results/` (guide, beginner, JSON, timestamped archives).

8. Interpretation, truthfulness, and limitations

- Reconstructed hotspots show where energy focuses under chosen geometry/beamformer — not pathology labels.
- Confidence scores are heuristic blends, not calibrated clinical probabilities.
- Uncertainty factors: antenna geometry, wave speed, clutter, ROI thresholds, limited healthy-vs-tumor calibration.
- Assumptions: BMID _adi uses pass-through preprocess; .s2p uses mild defaults; BMID automation picks first tumor else first scan.
- Known gap: GUI reconstruction still primarily uses filtered S21(f); Week-3 time-domain is stored/exported but not always default Module 3 input.

Model output is not a clinical diagnosis.

9. Roadmap

- Calibrate Module 8–10 and automation gates on larger healthy-vs-tumor BMID subsets.
- Optional Week-3 time-domain as default reconstruction input.
- Expand MCP tools for per-module dry-runs.
- Submission-ready PDF export of every automation run.

10. Key artifact locations

Artifact	Location
Session reports	project/results/latest_session_report.*
Guide report (A)	project/results/latest_guide_progress_report.md
Beginner report (B)	project/results/latest_beginner_explainer.md
Auto validation JSON	project/results/latest_auto_validation.json
Modules overview PDF	project/docs/Microwave_Modules_Full_Report.pdf
This detailed PDF	project/results/Microwave_Project_Detailed_Report.pdf
Auto validation docs	project/docs/AUTO_VALIDATION.md